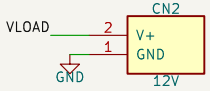
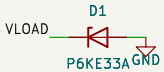
ESP32-S3-Zero Multiswitch: 8 Kanaele (ULN2803A) + CRSF – Entwurf v0.3

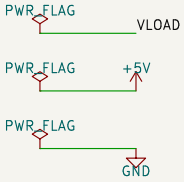
Last: max. 24V / 500mA je Kanal. ESP32 wird aus +5V von CN1 (BEC/Empfaenger) versorgt. Adresse per Firmware/Web-UI/Lua. Keine Pulldowns, keine UART-Vorwiderstaende. R9-R16 und R1-R8: Standard 150R, je Verbraucher waehlen.

UART A = zum Empfaenger bzw. vorigen Modul (Pin3 RX, Pin4 TX). UART B = zum naechsten Modul (Pin3 TX, Pin4 RX): 1:1-Kabel genuegt.

Lastversorgung 12-24V (VLOAD)







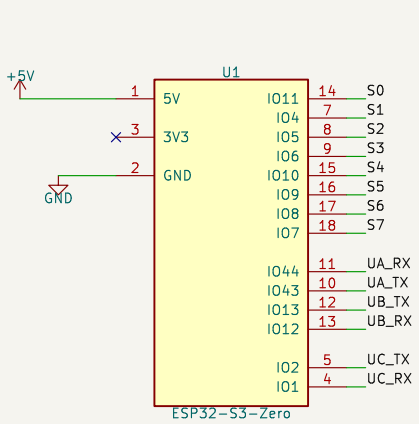
D1: TVS Anode GND, Kathode VLOAD (P6KE33A: ok bis 24V)

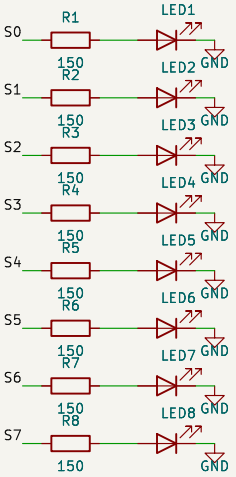
8-Kanal-Lastschalter (low-side), Eingaenge direkt vom ESP32

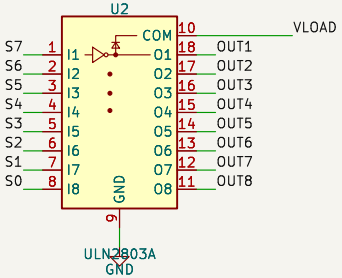
J1: ungerade Pins = OUT1..8 (ueber R9-R16), gerade Pins = VLOAD

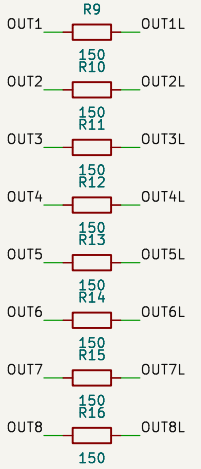
Kontroll-LEDs (Sn -> R -> LED -> GND)

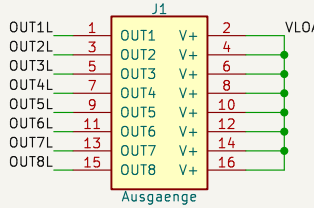
Ausgangswiderstaende (THT, je Verbraucher waehlen)



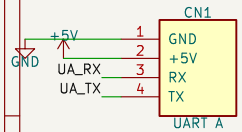




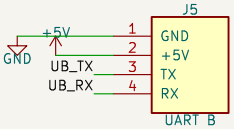




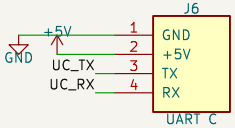
UART A (Empfaenger / vorheriges Modul)



UART B (naechstes Modul)



UART C (frei: S.Port o.ae.)



LED-Vorwiderstand R1-R8 bei 3,3V (ca. 5mA, mittlere Helligkeit):

Rot: 270R

Gelb: 270R

Gruen: 220R

Blau: 56R

Weiss: 56R

Entwurf – vor Layout/Bestellung pruefen

Sheet: /
File: CRSF-Multiswitch-ESP32-S3.kicad_sch

Title: CRSF Multiswitch ESP32-S3-Zero (ULN2803A)

Size: A4	Date: 2026-10-14	**Rev: v0.3**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1